## TABLE 1: COMPARISON OF THE REFERENCE DESIGNS

| Company | Module | Charge Management IC | No. of Cells | Suitable Chemistry | Max Charge Current (in amps) | Input Voltage (in volts) | Registration Required |
|---|---|---|---|---|---|---|---|
| Texas Instruments | TI TIDA-00590 | BQ25890/2 | 1S | Li-ion/Li-Po | 5A | 3.9-14 | No |
| Microchip | MCP1630 | MCP1630 | 1S or 2S | Li-ion/NiCD/NiMH | 2A | 10-28 | Yes |
| Monolithic Power | EV2639A-R-00B | MP2639A | 2S | Li-ion | 2.5A | 4.0-5.75 | No |
| ADI | ADI DC2703A-A-KIT | LT8491 | Multi-configu-ration | Li-ion, LiPo, SLA, etc | 16.6A (@12V) | 17-54 | No |
| Maxim Integrated | MAXREFDES1219 | MAX14746 | 1S | Li-ion | 3A | 2.2-5.5 | Yes |
| Renesas | DA9318 Reference board | DA9318 | 1S | Li-ion | 10A | 4.3-24 | No |

TIDA 00590 utilises two charging ICs marked as U1 and U2 in a dual IC configuration, which provides more than their 5A maximum charge current and distributes the heat loss across the board more efficiently. The charging IC bq25890/2 is a highly integrated 5A switch-mode battery charge management and system power path management device for single-cell Li-ion and Li-polymer batteries.

To achieve higher efficiency, the module offers a low impedance power path, which also reduces battery charging time while extending the battery life during discharging phase. The IC also integrates input current optimiser (ICO) and resistance compensation (IRCOMP) to deliver maximum charging power to the battery.

The module is over 90% efficient with a maximum efficiency of about 94%. The efficiency of the charger decreases as the charge current is increased.

This device supports various input sources, which include a standard USB host port, USB charging port, and USB compliant adjustable high voltage adaptor. Furthermore, it uses MaxCharge handshake using D+/D– pins and DSEL pin for USB switch control.

The device is compliant with USB 2.0 and USB 3.0 power specs with input current and voltage regulation. The charger features multiple safety features, including battery temperature negative thermistor monitoring, charging safety timer, overvoltage/overcurrent protections, undervoltage protection, and over-discharge protection.

The charger also employs a TS3USB221A, which is a high-bandwidth switch that requires a 3.3V supply. Since there are two charging ICs with a single USB I/O for the charger, both the power management ICs need to work in sync and need to transfer data simultaneously. Therefore, there's a need for a USB hub or controllers.

The TS3USB221A solution can effectively expand the limited USB I/O by switching between multiple USB buses to interface them to a single USB hub or controller. The switch works as a multiplexer; it takes a single USB input and gives output to both the bq25890/2 charging ICs. To provide a 3.3V output to the TS3USB221A, an LDO LP2985AIM5-3.3/NOPB is used.

***Important components.***

U1 - BQ25890RTWR charging management IC

U2 - BQ25892RTWR charging management IC

U3 - TS3USB221A high-bandwidth switch

U4 - LP2985AIM5 LDO

***Features.***

- Fast charges a smartphone on PCB with a low thermal budget
- Charge currents >5A
- This circuit design is tested and uses the EVM GUI and getting started guide
- High efficiency
- Integrated ADC
- Power path
- USB C/PD compatible
- USB D+/D-/BC1.2 integrated
- USB OTG integrated
- Operating voltage 3.9V-14V
- I2C communication
- LED indication for status signals
- Test points for key signals available for testing purposes
- Jumpers for different circuit configuration

***Protection features.***

- BAT temp thermistor monitoring (JEITA profile)
- BAT temp thermistor monitoring (hot/cold profile)
- IC thermal regulation
- Bad adaptor detection
- ICO (input current optimisation)
- IINDPM (input current limit)
- Input OVP

The module is suitable for use in mobile phones, cameras, tablets, standalone chargers, etc. You can find all the resources for TIDA 00590 by scanning the QR code.

### MCP1630 multi-bay charger for 2S configuration

The MCP1630 Li-ion charger is a versatile charger design capable of charging up to two single-cell Li-ion battery packs in a parallel configura-